

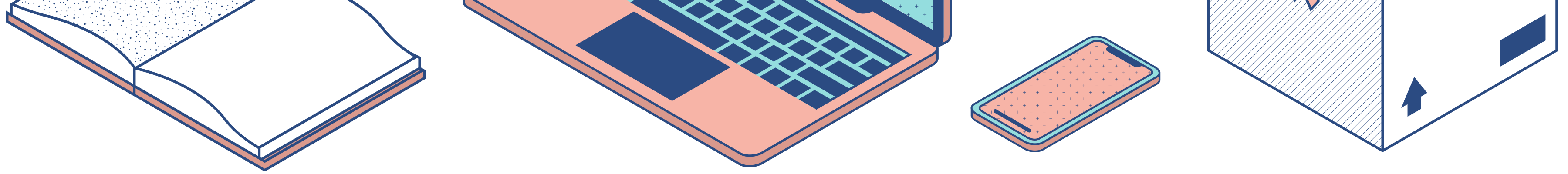
ATENEO DE ZAMBOANGA UNIVERSITY



TCP/IP Stack

ALEEKHAZER J. JAUSAN

CIT.016 | DATA COMMUNICATION



TCP (Transmission Control Protocol)

- Core protocol of the internet that defines and handles how data is transmitted over networks
- Manages data transmission between devices, ensuring reliable, ordered, and error-checked delivery of data between applications

IP (Internet Protocol)

- provides addressing and routing of packets across networks
- routing packets of data across networks from source to destination using IP addresses as the basis





TCP/IP STACK

1 ————— 2 ————— 3 ————— 4

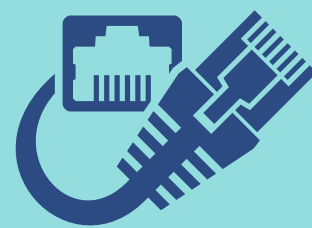
Application Layer

- provides network services directly to end-user applications
- web browsing, email, and file transfers



Transport Layer (TCP)

- manages end-to-end communication between devices
- data integrity, error recovery, and flow control



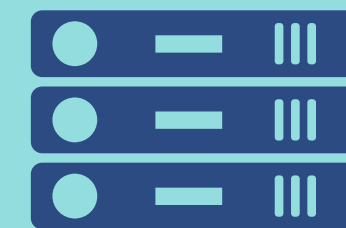
Internet/Network Layer (IP)

- logical transmission of data over the entire network
- packets are delivered accurately to the right directions using IP addresses



Network Access / Data Link Layer

- a group of applications requiring network communications
- responsible for generating the data and requesting connections
- error prevention and "framing"

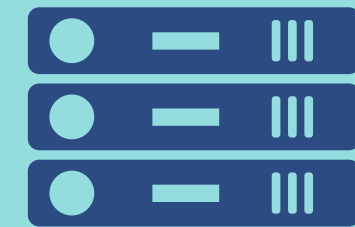
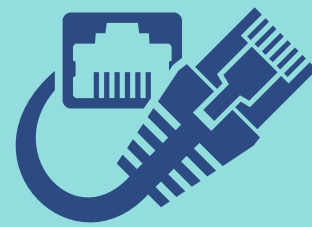




During the icebreaker, what
part of the game was the:

APPLICATION LAYER

1 ————— 2 ————— 3 ————— 4

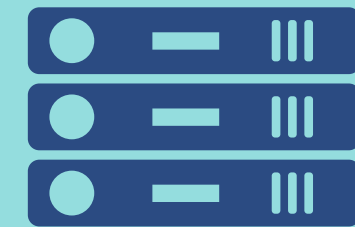
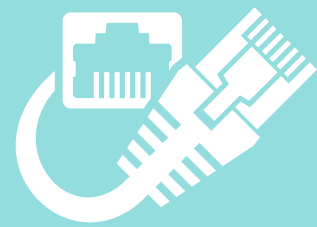




During the icebreaker, what
part of the game was the:

TRANSPORT LAYER

1 ————— 2 ————— 3 ————— 4

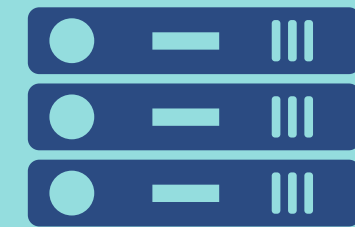
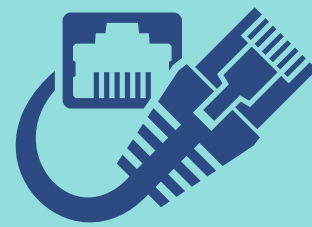




During the icebreaker, what
part of the game was the:

INTERNET LAYER

1 ————— 2 ————— 3 ————— 4

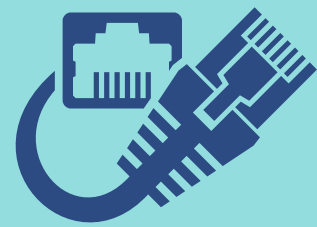


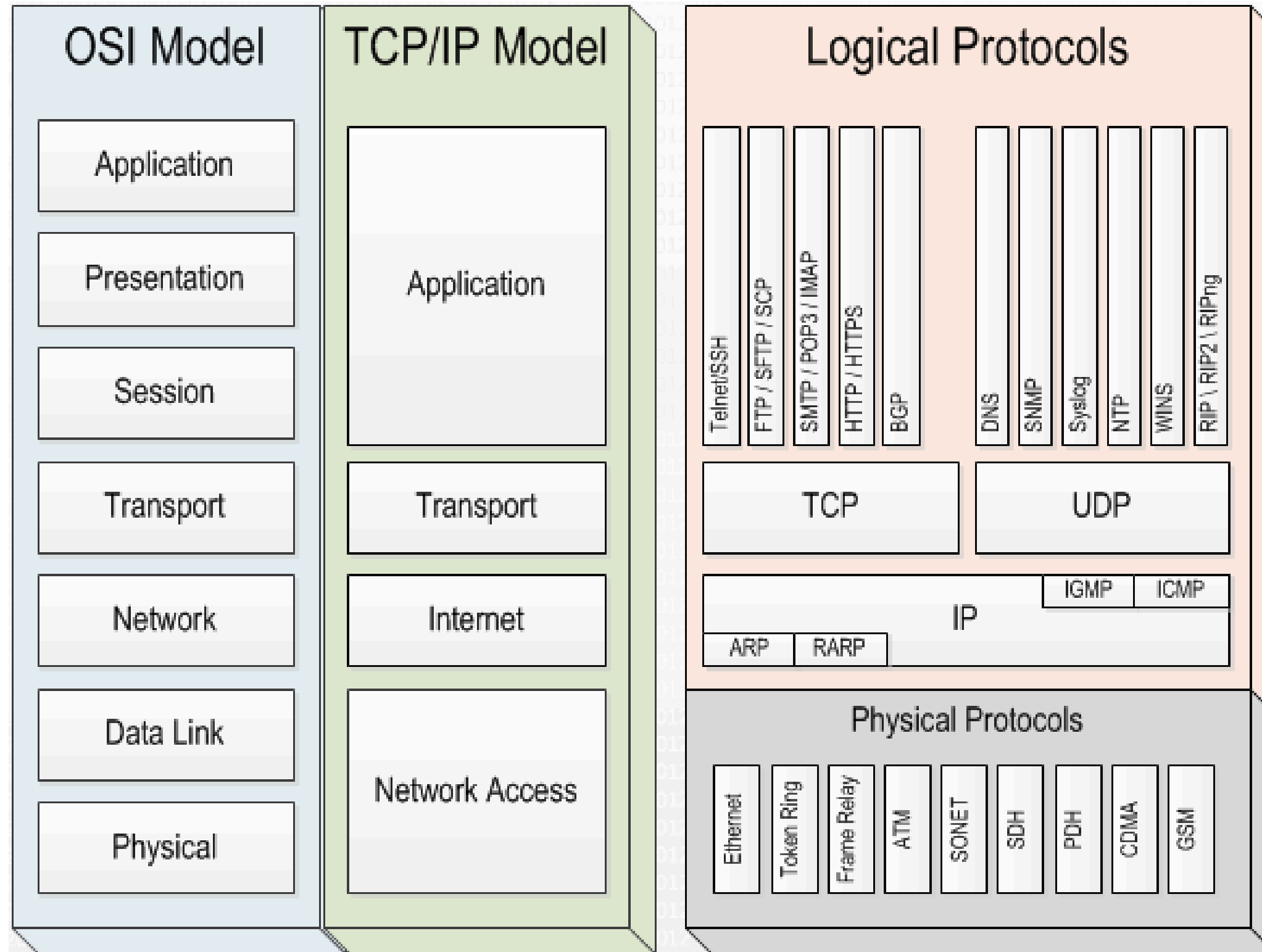


During the icebreaker, what
part of the game is the:

DATA LINK LAYER

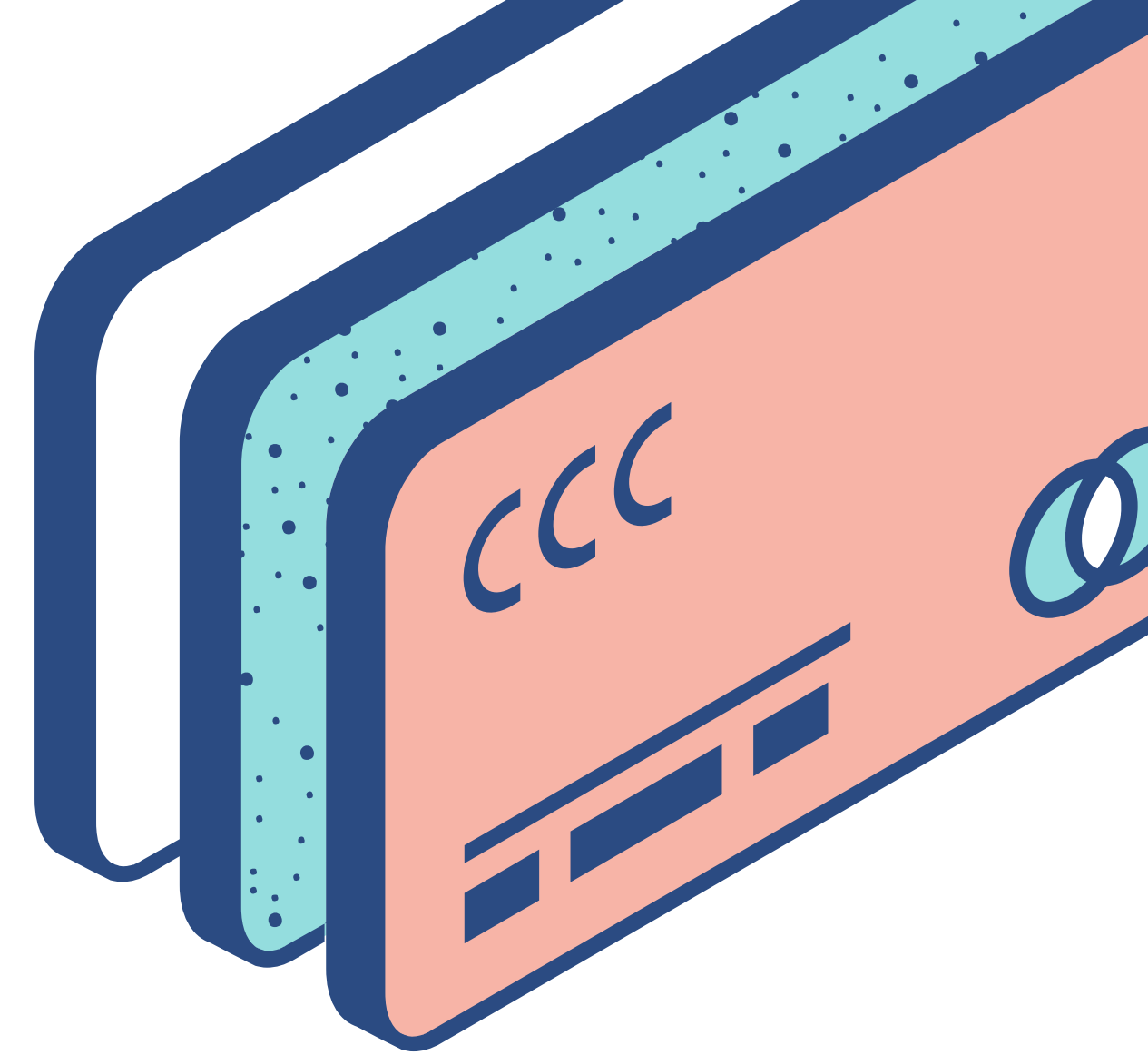
1 ————— 2 ————— 3 ————— 4





TCP/IP vs OSI

DIFFERENCE IN LAYERS





	TCP/IP (Transmission Control Protocol / Internet Protocol)	Open Systems Interconnection
Layers	Combined session and presentation layer in the application layer	Different session and presentation layers
Approach	Horizontal, connectionless	Vertical
Assurance delivery of packets	Does not guarantee	Guaranteed
Protocol replaceability	Difficult to replace	Better covered and are easy to replace
Network Layer	Connectionless (IP) services, transport layer (TCP) provides connections	Connectionless and connection-oriented services



TCP/IP vs OSI

DIFFERENCE



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THANK YOU

FOR LISTENING

